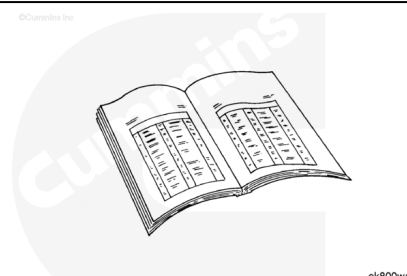


Trainings ▾

Preparatory Steps

Industrial and Power Generation

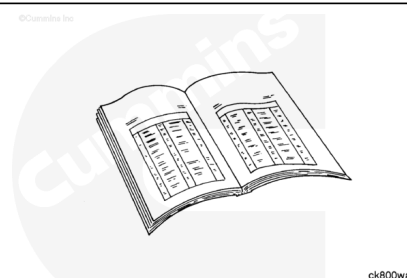
- Remove the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Remove the rocker lever assembly. Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html).
- Remove the push rods. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Remove the crossheads. Refer to Procedure 002-001 (/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html).
- Remove the crankcase breather. Refer to Procedure 003-001 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).
- Remove the wiring harness. Refer to 019-043 (/qs3/pubsys2/xml/en/procedures/56/56-019-043-tr.html).
- Remove the cam follower cover. Refer to Procedure 004-002 (/qs3/pubsys2/xml/en/procedures/56/56-004-002-tr.html).



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QSK60 Marine Applications

- Remove the aftercooler heat shields. Refer to Procedure 010-129 (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html).
- Remove the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Mark the location of the rocker lever assemblies as they are removed, so they can be installed in the original location.
- Remove the rocker lever assembly. Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/56/56-003-



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009-tr.html).

- Mark the push rods so they can be reinstalled in their original position.
- Remove the push rods. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Remove the crossheads.
- When removing the cam follower assemblies located on the left bank of the engine, the crankcase breather **must** be removed. Refer to Procedure 003-001 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html) for procedures to remove the crankcase breather.
- When removing the cam follower assemblies located on the left bank of the engine, the wiring harness and wiring harness bracket **must** be removed. Refer to 019-043 (/qs3/pubsys2/xml/en/procedures/19/19-019-043.html) in the Troubleshooting and Repair Manual, QSK Fuel Systems, QSK19, QSK23, QSK45, QSK60, and QSK78 Series Engines, Bulletin 3666113 for procedures to remove the wiring harness.
- Remove the cam follower cover. Refer to Procedure 004-002 (/qs3/pubsys2/xml/en/procedures/56/56-004-002-tr.html).

with Electronically Actuated Injector

- Remove the injector supply lines. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/56/56-006-024-tr.html).
- Remove the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Remove the rocker lever assembly. Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html).
- Remove the push rods. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Remove the left bank and right bank wiring harness. Refer to 019-043 (/qs3/pubsys2/xml/en/procedures/56/56-019-043-tr.html).
- Remove the crankcase breather. Refer to Procedure 003-001 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).



- Remove the fuel return manifold. Refer to Procedure 006-021 (/qs3/pubsys2/xml/en/procedures/56/56-006-021-tr.html).
- Remove the fuel pump. Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html).
- Remove the cam follower cover. Refer to Procedure 004-002 (/qs3/pubsys2/xml/en/procedures/56/56-004-002-tr.html).

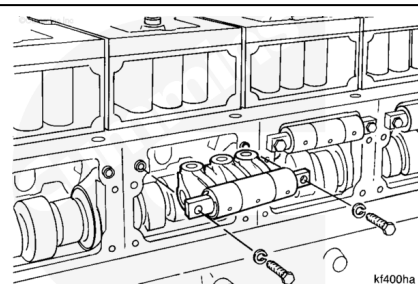
Remove

with Mechanically Actuated Injector

Mark the location of the cam follower assemblies as they are removed, so they can be installed in the original location.

Remove the cam follower mounting capscrews.

Pull the cam follower assembly straight out until the assembly is off the ring dowels.

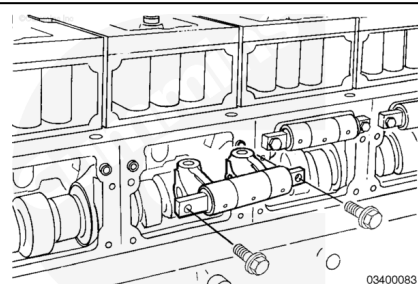


with Electronically Actuated Injector

Mark the location of the cam follower assemblies as they are removed, so they can be installed in the original location.

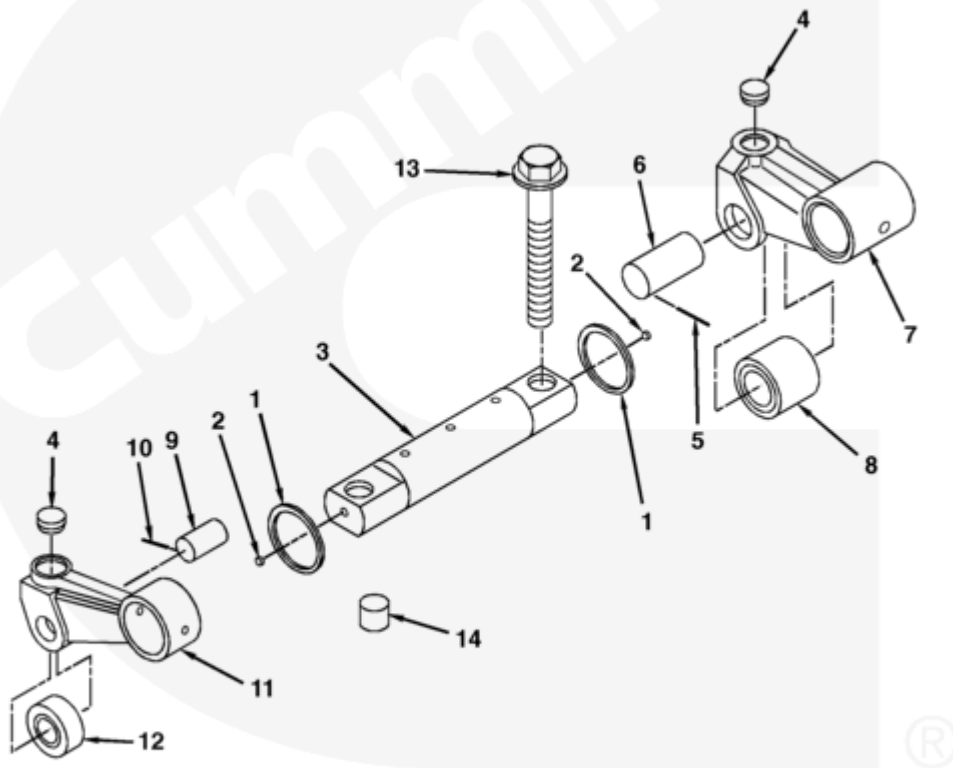
Remove the two capscrews and washers securing the cam follower assembly.

Pull the cam follower assembly straight out until the assembly is off the ring dowels.



Exploded View

with Mechanically Actuated Injector

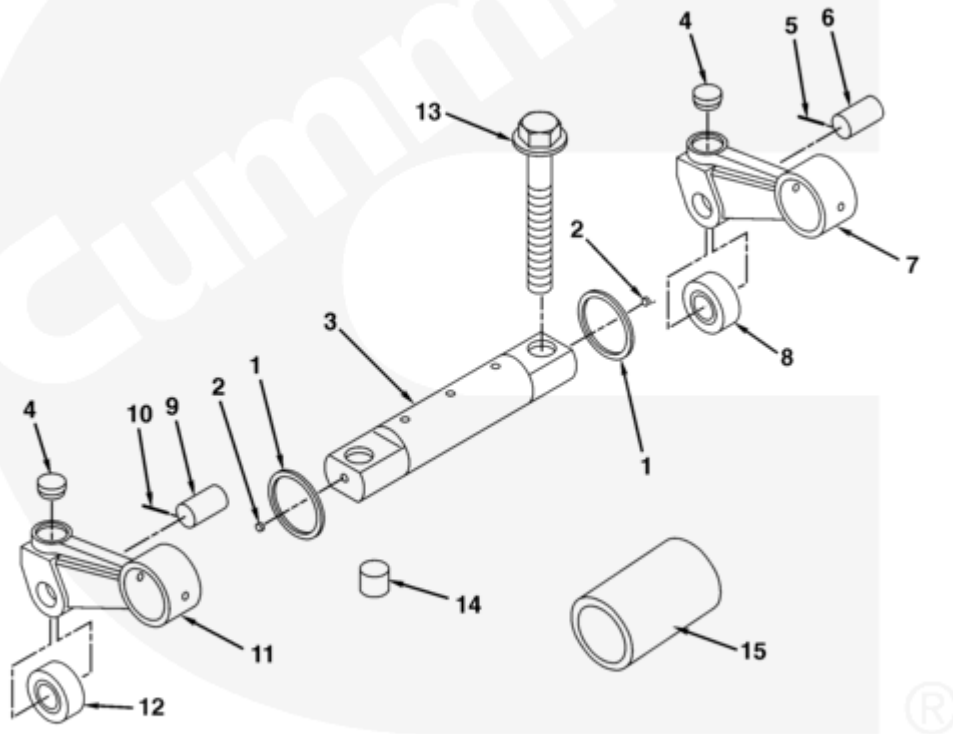


04800006

1. Retaining ring
2. Expansion plug
3. Cam follower shaft
4. Cam follower socket
5. Tappet roller pin lockwire
6. Cam follower roller pin
7. Injector follower lever
8. Cam follower roller
9. Cam follower roller pin
10. Tappet roller pin lockwire
11. Valve follower lever
12. Cam follower roller
13. Hexagon head flange capscrew
14. Dowel ring.

Note : The cam follower roller pin (6) and the cam follower roller (8) are not available for service.

with Electronically Actuated Injector



04600028

1. Retaining ring
2. Expansion plug
3. Cam follower shaft
4. Cam follower socket
5. Tappet roller pin lockwire
6. Cam follower roller pin
7. Valve follower lever
8. Cam follower roller
9. Cam follower roller pin
10. Tappet roller pin lockwire
11. Valve follower lever
12. Cam follower roller
13. Hexagon head flange capscrew
14. Dowel ring
15. Spacer.

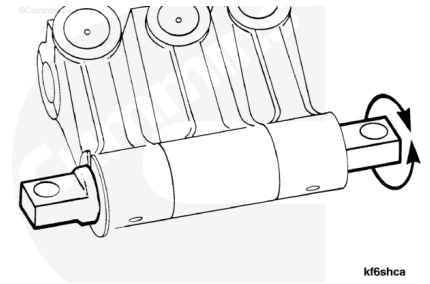
Note : The cam follower roller pin (6) and the cam follower roller (8) are not available for service.

Disassemble

with Mechanically Actuated Injector

If the engine has excessive wear because of air cleaner or air inlet tube failure, the cam followers **must** be replaced.

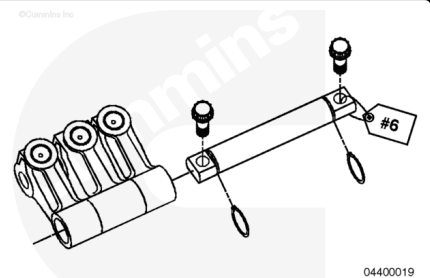
The shaft **must** rotate easily. If resistance is apparent, disassemble and check for burrs on the shaft.



Remove the 12 point capscrews.

Remove the retaining ring and pull the shaft out.

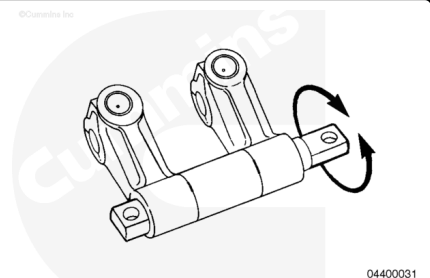
Mark and tag the parts for future installation.



with Electronically Actuated Injector

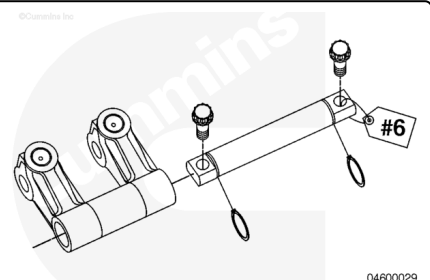
If the engine has excessive wear because of air cleaner or air inlet tube failure, the cam followers **must** be replaced.

The shaft **must** rotate easily. If resistance is apparent, disassemble and check for burrs on the shaft.



Remove the retaining ring and pull the shaft out.

Mark and tag the parts for future installation.

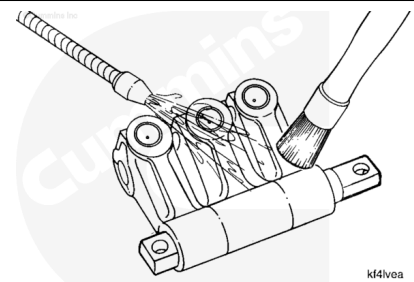


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



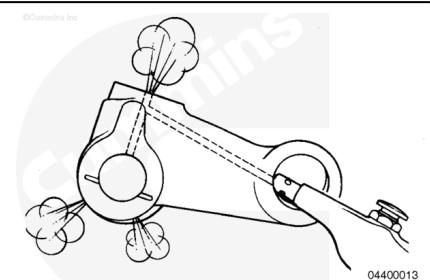
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the cam follower assembly with solvent and dry with compressed air.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the cam followers with solvent and dry with compressed air.

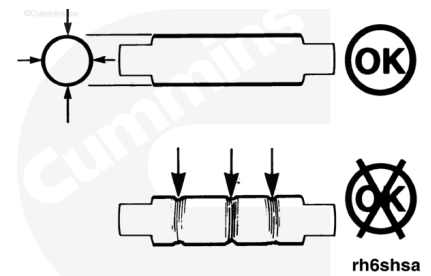
Use compressed air to blow through the oil drillings making sure they are **not** plugged.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Clean the cam follower shaft with solvent and dry with compressed air.

Use compressed air to blow through the nine oil drillings of the shaft making sure they are **not** plugged.

Inspect the cam follower shaft for damage and roughness.

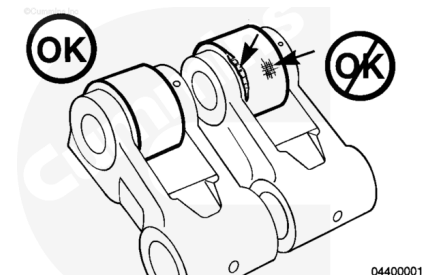
Measure the cam follower shaft outside diameter.

mm		in
39.975	MIN	1.574
40.013	MAX	1.575

Replace the cam follower shaft if it is **not** within specifications.

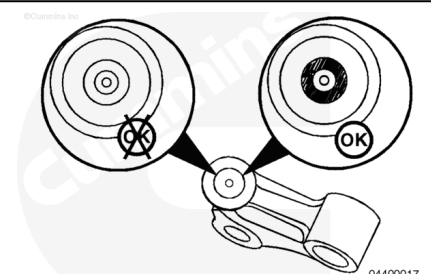
Inspect the cam follower roller for damage.

Replace the cam follower assembly if the roller is damaged.



Inspect the cam follower push rod socket for damage.

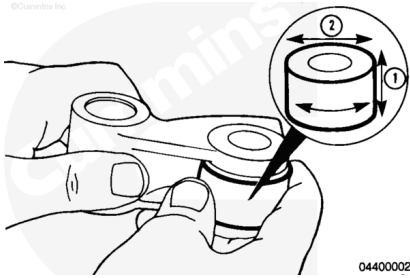
Replace the cam follower if the push rod socket is damaged.



Rotate the roller it **must** rotate easily.

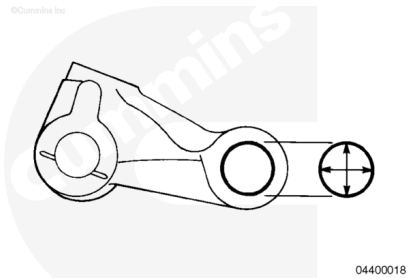
Measure the roller clearance.

	mm		in
Reference Point 1	0.230	MIN	0.009
	0.610	MAX	0.024
Reference Point 2	0.076	MIN	0.003
	0.114	MAX	0.005



Measure the cam follower shaft bore inside diameter.

mm		in
40.023	MIN	1.576
40.073	MAX	1.578

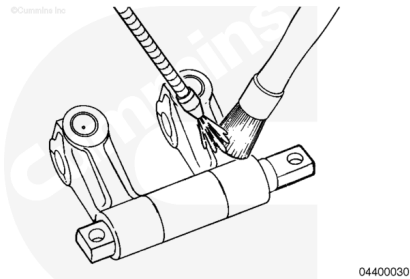


If the cam follower lever inside diameter does **not** meet specifications the cam follower lever assembly **must** be replaced.

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the cam follower assembly with solvent and dry with compressed air.

⚠ WARNING ⚠

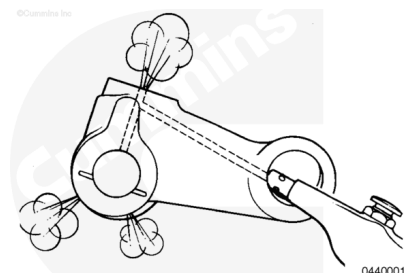
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

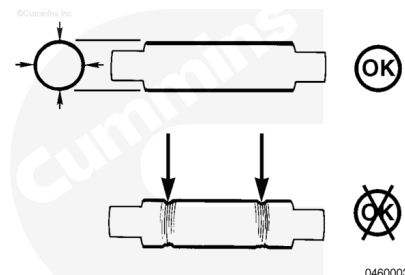
Clean the cam followers with solvent and dry with compressed air.

Use compressed air to blow through the oil drillings making sure they are **not** plugged.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the cam follower shaft with solvent and dry with compressed air.

Use compressed air to blow through the nine oil drillings of the shaft making sure they are **not** plugged.

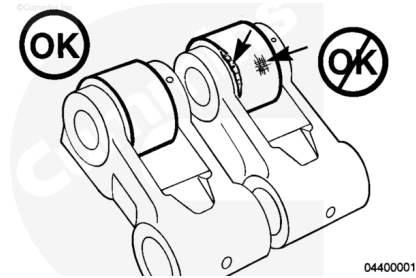
Inspect the cam follower shaft for damage and roughness.

Measure the cam follower shaft outside diameter.

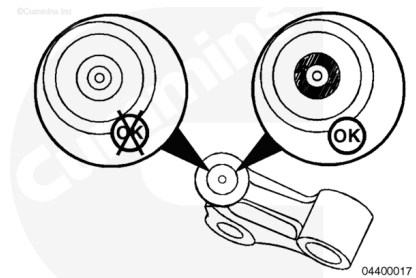
mm		in
39.975	MIN	1.574
40.013	MAX	1.575

Replace the cam follower shaft if it is **not** within specifications.

Inspect the cam follower for reuse. The cam follower lever assembly **must** be replaced if damaged.

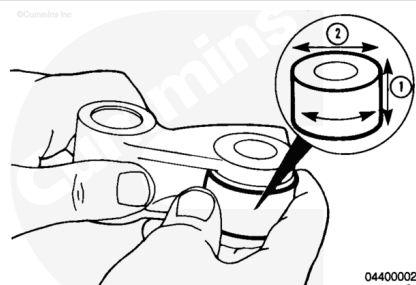


Inspect the cam follower push rod socket for damage.
Replace the cam follower if the push rod socket is damaged.



⚠ CAUTION ⚠

Do not touch the machined surfaces of the camshaft follower roller with bare hands. This will cause rust to form; before handling, lubricate the camshaft follower roller with clean 15W-40 oil.



Rotate the roller, it **must** rotate easily.

Measure the clearance of the roller.

	mm		in
Reference Point 1	0.230	MIN	0.009
	0.610	MAX	0.024
Reference Point 2	0.076	MIN	0.003
	0.114	MAX	0.005

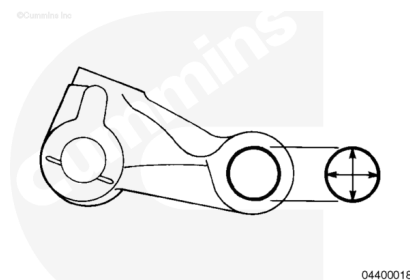
If the clearances do **not** meet specifications, the cam follower lever assembly **must** be replaced.

Measure the cam follower shaft bore inside diameter.

mm		in
40.023	MIN	1.576

mm		in
40.073	MAX	1.578

If the cam follower lever inside diameter does **not** meet specifications the cam follower lever assembly **must** be replaced.



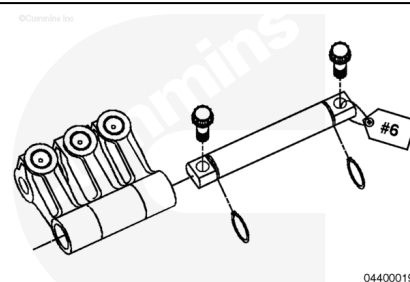
Assemble

with Mechanically Actuated Injector

Use clean engine oil to lubricate the shaft and bores of the levers.

Slide the levers on the shaft as shown.

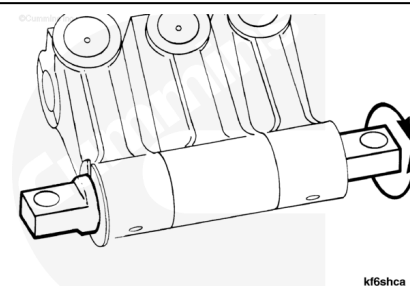
Install the retaining rings and capscrews.



Rotate the shaft.

The shaft **must** rotate freely after assembly.

If resistance is apparent, disassemble and check for burrs on the shaft.

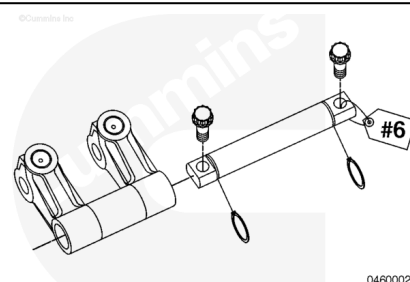


with Electronically Actuated Injector

Use clean engine oil to lubricate the shaft and bores of the levers.

Slide the levers on the shaft as shown.

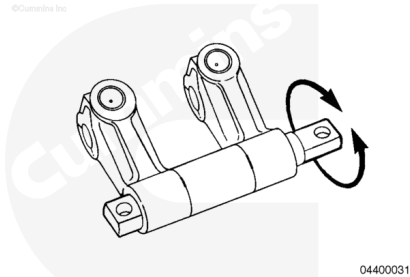
Install the retaining rings and capscrews.



Rotate the shaft.

The shaft **must** rotate freely after assembly.

If resistance is apparent, disassemble and check for burrs on the shaft.

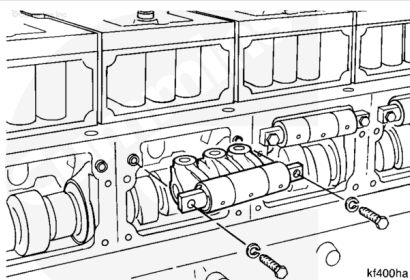


Install

with Mechanically Actuated Injector

⚠ CAUTION ⚠

Do not touch the machined surfaces of the camshaft follower roller with bare hands. This will cause rust to form; before handling, lubricate the camshaft follower roller with clean engine oil.



Use clean engine oil to lubricate the camshaft and cam followers.

Install the cam follower assembly. The shaft **must** fit on both ring dowels.

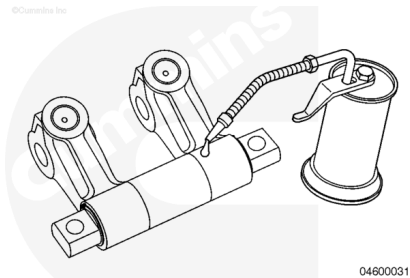
Install and tighten the mounting capscrews.

Torque Value:	Step 1	35 n.m [26 ft-lb]
	Step 2	140 n.m [103 ft-lb]
	Step 3	280 n.m [207 ft-lb]

with Electronically Actuated Injector

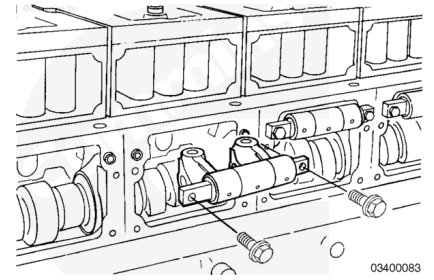
Note : Service replacement cam followers are coated with a heavy preservative to prevent rust. This preservative **must** be removed completely with solvent before the parts are installed on the engine. Use clean engine oil to lubricate the cam follower assembly and camshaft.

Lubricate the cam follower assembly.



⚠ CAUTION ⚠

Do not touch the machined surfaces of the camshaft follower roller with bare hands. This will cause rust to form; before handling, lubricate the camshaft follower roller with clean engine oil.



Install the cam follower assembly. The shaft **must** fit on both ring dowels.

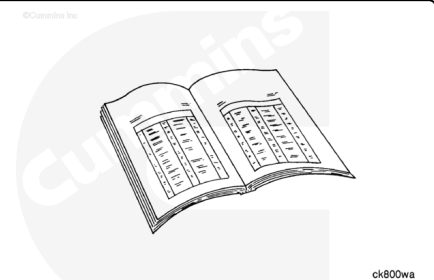
Lubricate the threads and install and tighten the mounting capscrews.

Torque Value:	Step 1	35 n.m [26 ft-lb]
	Step 2	140 n.m [103 ft-lb]
	Step 3	280 n.m [207 ft-lb]

Finishing Steps

Industrial and Power Generation

- Install the cam follower cover. Refer to Procedure 004-002 (</qs3/pubsys2/xml/en/procedures/56/56-004-002-tr.html>).
- Install the wiring harness. Refer to 019-043 (</qs3/pubsys2/xml/en/procedures/56/56-019-043-tr.html>).
- Install the crankcase breather. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html>).
- Install the push rods. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html>).
- Install the crossheads. Refer to Procedure 002-001 (</qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html>).
- Install the rocker lever assembly. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html>).
- Adjust the overhead. Refer to Procedure 003-006 (</qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html>).



tr.html).

- Install the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).

QSK60 Marine Applications

- Install the cam follower cover. Refer to Procedure 004-002 (/qs3/pubsys2/xml/en/procedures/56/56-004-002-tr.html).
- If the cam follower assembly located on the left bank of the engine was removed, install the wiring harness and harness brackets. Refer to 019-043 (/qs3/pubsys2/xml/en/procedures/19/19-019-043.html) in the Troubleshooting and Repair Manual, QSK Fuel Systems, QSK19, QSK23, QSK45, QSK60, and QSK78 Series Engines, Bulletin 3666113.
- If the cam follower assembly removed was located on the left bank of the engine, install the crankcase breather assembly. Refer to Procedure 003-001 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).
- Install the push rods. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Install the crossheads.
- Install the rocker lever assembly. Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html).
- Adjust the overhead using the outer base circle method. Refer to Procedure 003-006 (/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html).
- Install the rocker lever cover and all related parts. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Install the aftercooler heat shields. Refer to Procedure 010-129 (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html).

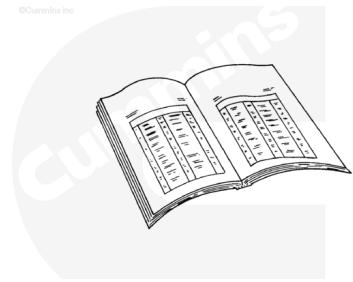


with Electronically Actuated Injector

- Install the cam follower cover. Refer to Procedure 004-002 (/qs3/pubsys2/xml/en/procedures/56/56-004-002-

tr.html).

- Install the left bank and right bank wiring harness. Refer to 019-043 (/qs3/pubsys2/xml/en/procedures/56/56-019-043-tr.html).
- Install the fuel pump. Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html).
- Install the fuel return manifold. Refer to Procedure 006-021 (/qs3/pubsys2/xml/en/procedures/56/56-006-021-tr.html).
- Install the crankcase breather. Refer to Procedure 003-001 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).
- Install the push rods. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html).
- Install the crossheads. Refer to Procedure 002-001 (/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html).
- Install the rocker lever assembly. Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html).
- Adjust the overhead. Refer to Procedure 003-006 (/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html).
- Install the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Install the injector supply lines. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/56/56-006-024-tr.html).



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